simon-5501-10-slides

2026-10-27

Topics to be covered

- What you will learn
  - The analysis of variance model
  - The F-test
  - Assumptions
  - Confidence intervals
  - R code for analysis of variance
  - Sample size justification
  - R code for sample size justification

Review two-sample t-test

- $H_0: \mu_1 = \mu_2$
- $H_1: \mu_1 \neq \mu_2$
- $T = \frac{\bar{X}_1 - \bar{X}_2}{se}$
  - Accept H0 if T is close to zero.

Speaker notes

You saw how to compare two means last week. Here is the general framework. In the earlier lectures, I used the null hypothesis H0 mu1-mu2=0, but it is often written in an equivalent form H0 mu1 = mu2.

What to do with three or more groups?

- Analysis of variance test framework
  - $H_0: \ \mu_1 = \mu_2 = \ldots = \mu_k$
  - $H_1: \ \mu_i \neq \mu_j$ for some i, j
  - Accept H0 if the F ratio (defined below) is close to 1.
- The alternative is NOT $H_0: \ \mu_1 \neq \mu_2 \neq \ldots \neq \mu_k$
  - Some equalities might be mixed in
  - Example: $\mu_1 = \mu_2$, but $\mu_1 \neq \mu 3$ and $\mu 2 \neq \mu 3$.

Speaker notes

With three or more groups, the hypothesis changes slightly. The null hypothesis is what you would expect, but the alternative is that at least one pair (possibly more than one pair) of means are not equal. So, for example, the population mean in the first group equals the population mean in the second group, but both of these are not equal to the population mean of the third group.

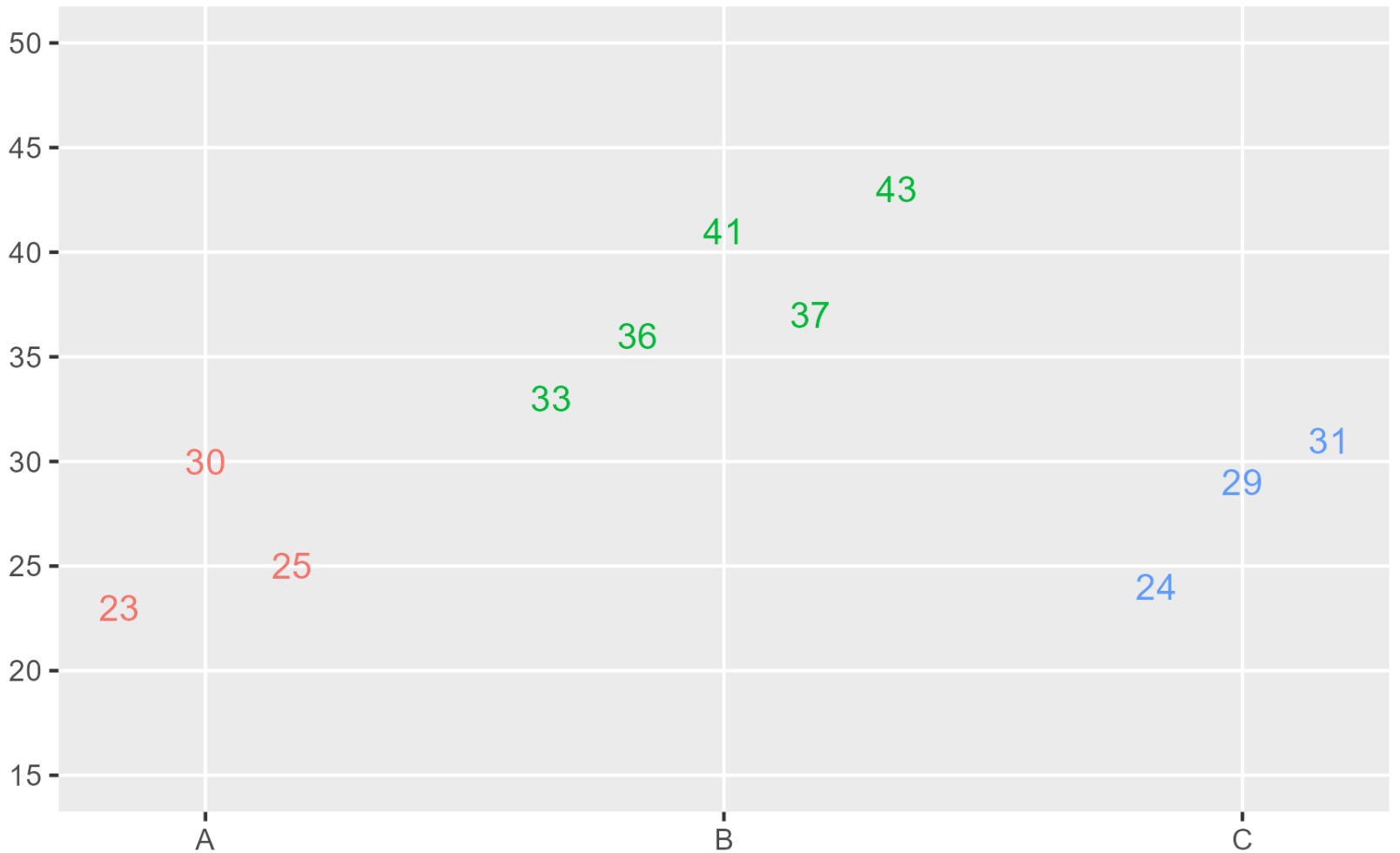
Artificial data

```
# A tibble: 11 × 2
   g         y
   <chr> <dbl>
 1 A        23
 2 A        30
 3 A        25
 4 B        33
 5 B        36
 6 B        41
 7 B        37
 8 B        43
 9 C        24
10 C        29
11 C        31
```

Speaker notes

To motivate some of the calculations in Analysis of Variance, I created an artificial data set with numbers that are easy to work with.

Scatterplot

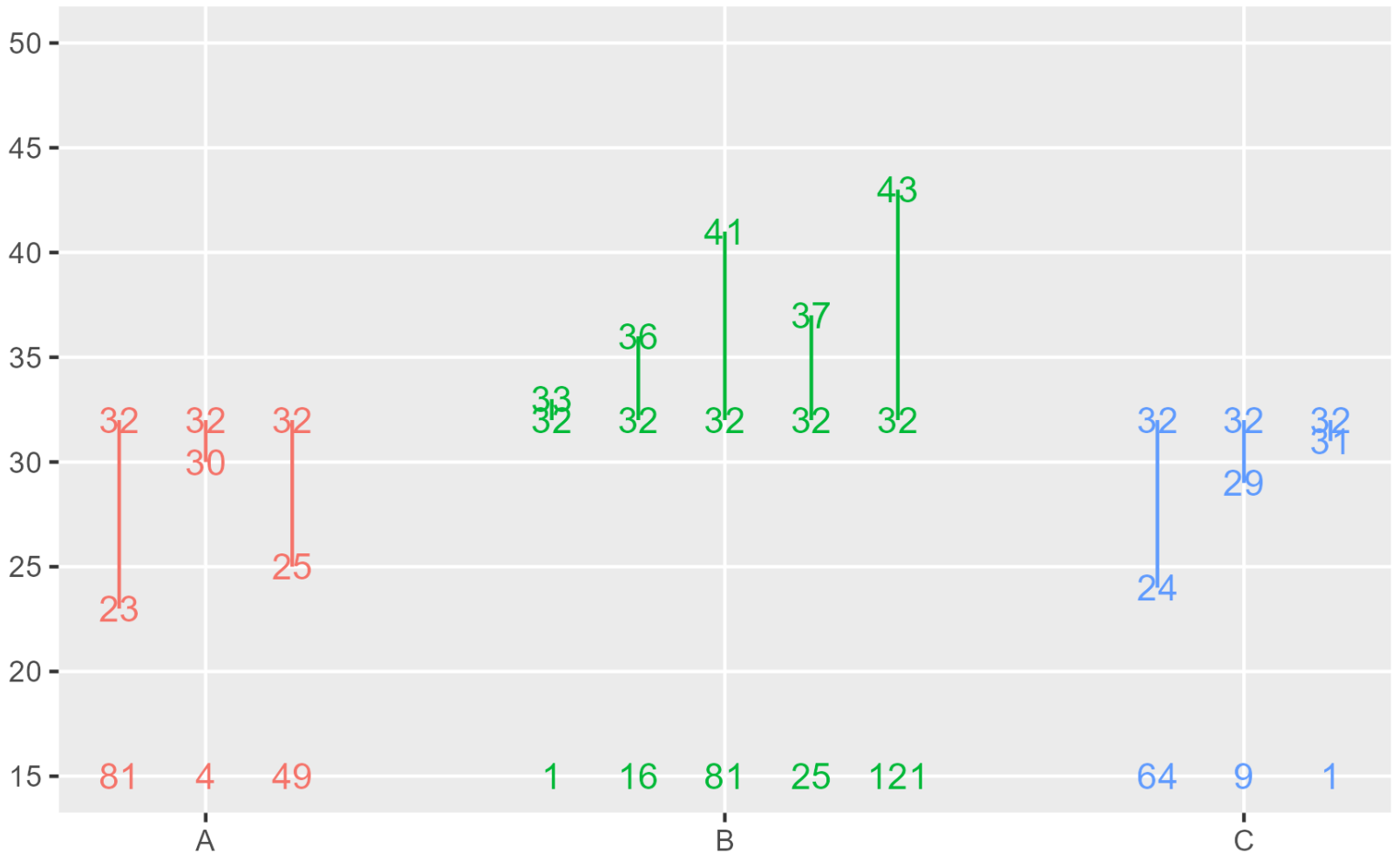


Simon, 2024-10-22

Speaker notes

Here is a plot of the data.

SS(Total) or SST = 452

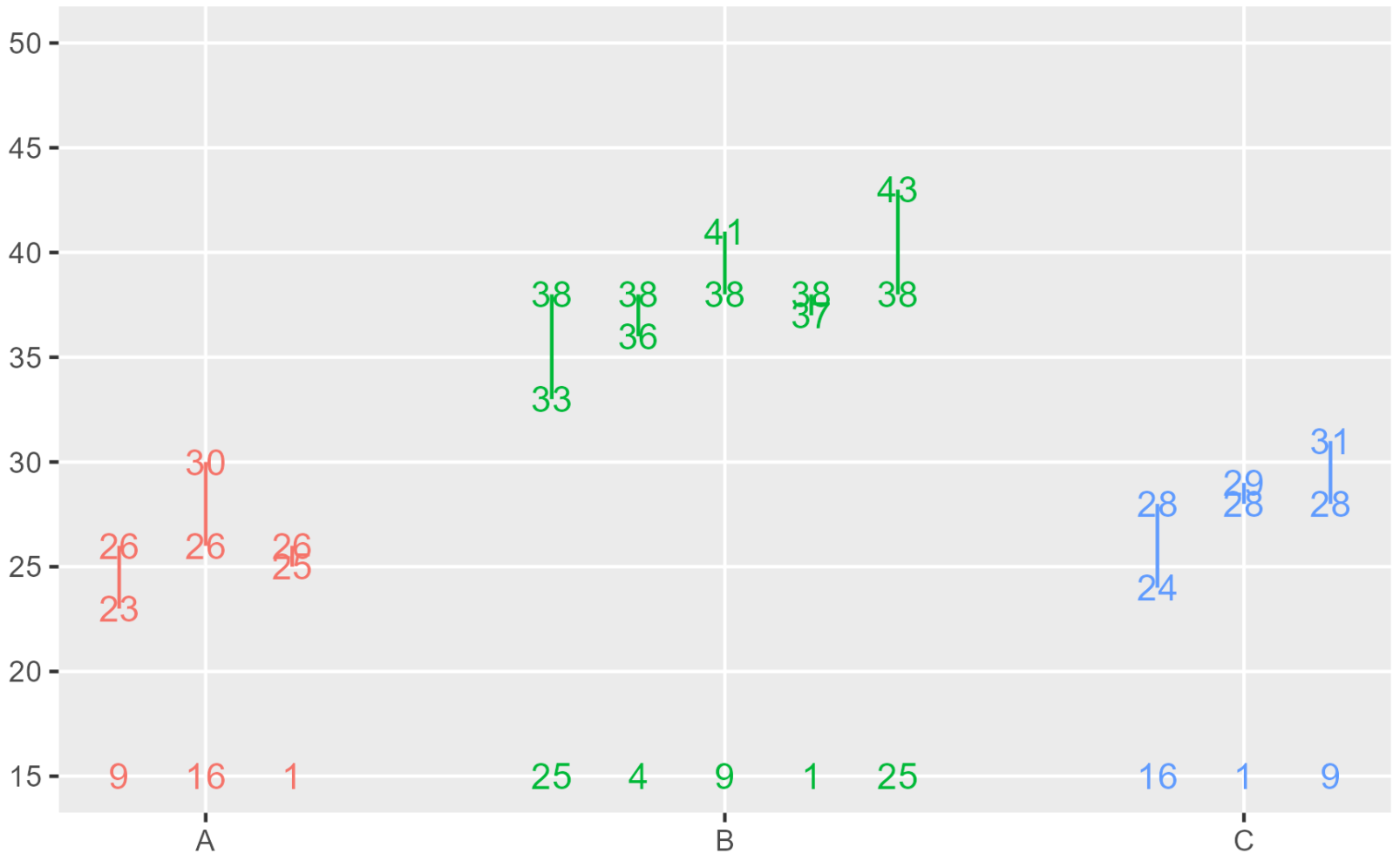


Simon, 2024-10-22

Speaker notes

Total sum of squares is the squared deviation between each individual value and the overall mean. The first observation is 23, 9 units below the overall mean of 32. This contributes 81 to the total sum of squares. The second observation is 30, 2 units below the overall mean of 32. This contributes 4 to the total sum of squares. Keep on going. The last observation is 31, 1 unit below the overall mean of 32. It contributes 1 to the total sum of squares.

SS(Within) or SSW = 116

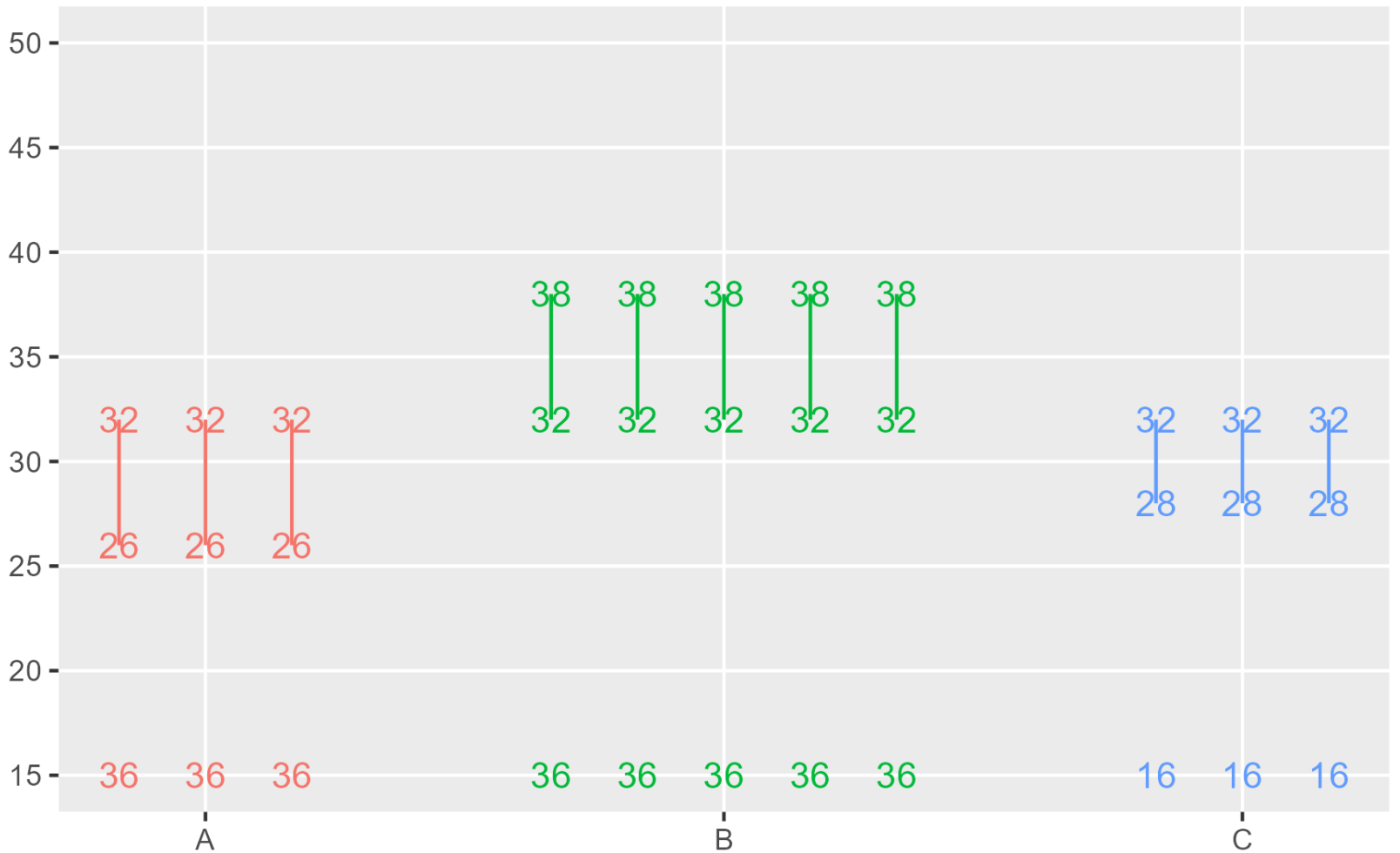


Simon, 2024-10-22

Speaker notes

Within sum of squares is the squared deviation between each individual value and the group means. The first group has a mean of 26. The first observation in the group, 23, is 3 units below the group 1 mean and contributes 9 to the within sum of squares. The second observation in the group, 40, is 4 units above the mean and contributes 16 to the within sum of squares. The third observation in the group, 25, is one unit below the group 1 mean and contributes only 1 to the within sum of squares. You compare the fourth observation, 33, to the group 2 mean. It is 5 units below the group 2 mean and contributes 25 to the within sum of squares. And so on.

SS(Between) or SSB = 336



Simon, 2024-10-22

Speaker notes

The between sum of squares is the squared deviation between the group means and the overall mean. In the first group, there are three observations. In each case, the group 1 mean, 26, is 6 units smaller than the overall mean. So the first three observations each contribute 36 to the between sum of squares. The next five observations come from a group with a mean of 38. This is 6 units above the overall mean, so each of these five observations contribute 36 to the between sum of squares. Finally, the last three observations have a group mean of 28, which is 4 units below the overall mean. So the last three observations contribute 16 to the between sum of squares.

Interpretation

- SST is the total variation in the dataset.
- SSE is the unexplained variation in the dataset
- SSB is the explained variation in the dataset
  - $R^2 = \frac{SSB}{SST}$ is the proportion of explained variation

Speaker notes

The sum of squares total represents the variation about the overall mean. It can be partitioned into unexplained variation (sum of squares error) and explained variation (sum of squares between). The ratio of sum of squares between to sum of squares total is denoted by R-squared and has the same interpretation that you saw in linear regression, the proportion of explained variation.

If sum of squares between is small, there is very little variation in the group means. Thois means that knowing group membership does not help that much in predicting your outcome. if the sum of squares between is large, there is quite a bit of variation in the group means. One mean might be really big, another really small and a third in the middle. Knowing which group a patient is in tells you a lot about whether the outcome would be big, small, or somewhere in the middle.

Other names for the sum of squares

- SSW is sometimes abbreviated as SSE or SSR
- SSB is sometimes abbreviated as SSR(!)

Speaker notes

There is some ambiguity in how these sums of squares are labeled. You may see SSW abbreviated as SSE (E stands for error) or SSR (R stands for residual). You may see SSB abbreviates as SSR (R stands for regression). Be careful, of course and don’t abbreviate both sums of squares as SSR.

Degrees of freedom

- For SS(Total), df = 10
- For SS(Within), df = 8
- For SS(Between), df = 2
- In general,
  - n = number of observations total
  - k = number of groups
    - Total df = n-1
    - Within df = n-k
    - Between df = k-1

Speaker notes

The concept of degrees of freedom is tricky. It is the number of “independent” observations, or the number of observations minus the number of estimated parameters.

For Total SS, you have 11 observations, but one estimated parameter, the overall mean of 32. The degrees of freedom is 11-1 = 10.

For Within SS, you also have 11 observations, but there are 3 estimated parameters, the three group means. The degrees of freedom is 11-3 = 8

For Between SS, you only have three observations, the three group means. There is one estimated parameter, the overall mean. The degrees of freedom is 3-1 = 2.

In general, if you let n represent the total number of observations across all groups and let k represent the number of groups, then the degrees of freedom are n-1, n-k, and k-1.

R calculations of sums of squares

```
       term df sumsq meansq
1 factor(g)  2   336  168.0
2 Residuals  8   116   14.5
```

Speaker notes

Here is the analysis of variance table for the artificial dataset as computed by R. I have left out the F-ratio and the p-value, which you will hear about in the next video.

Break #1

- What you have learned
  - The analysis of variance model
- What’s coming next
  - The F-test

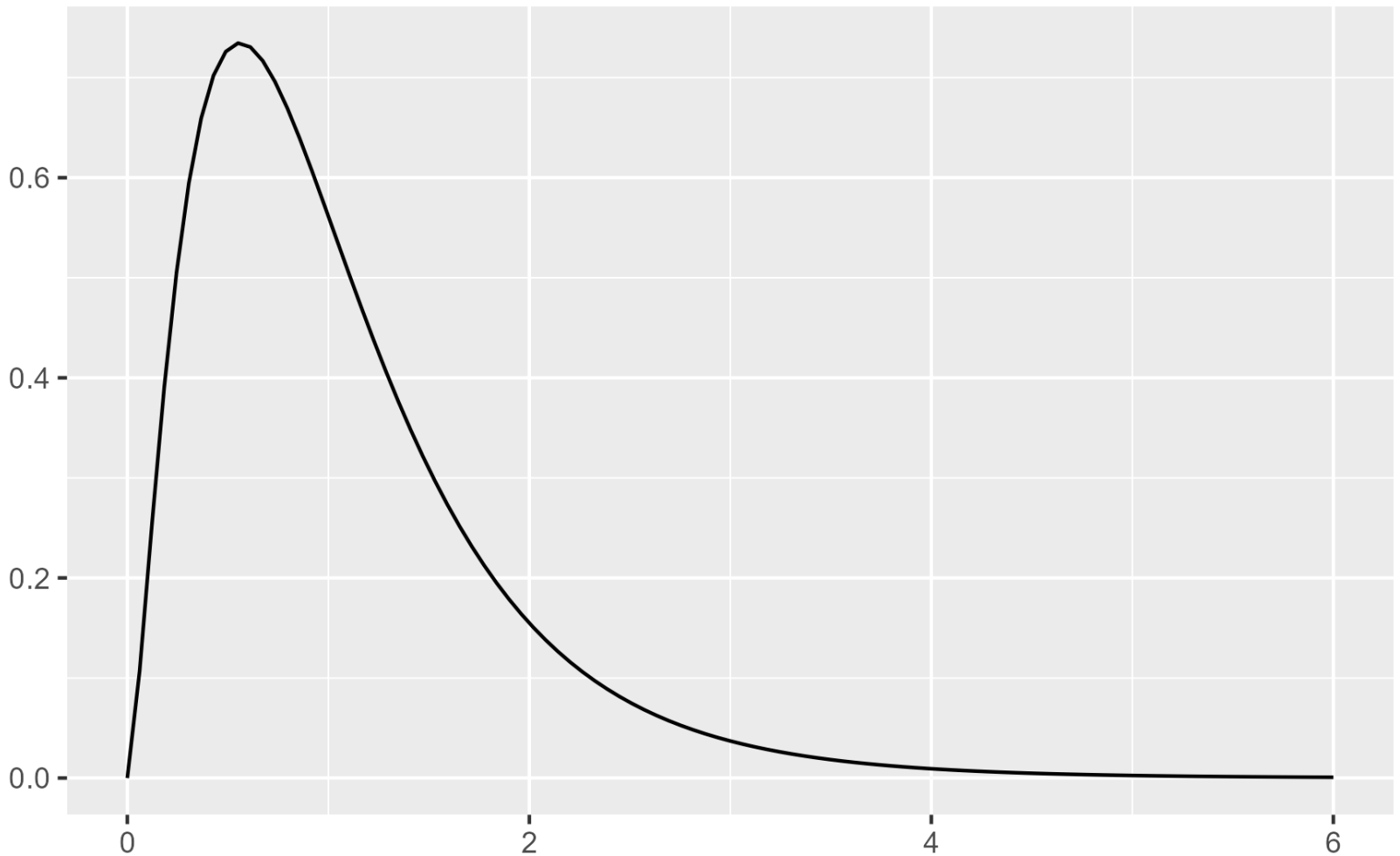
The F-distribution

- Ratio of two measures of variation
- Two measures are comparable if F is close to 1
- F distribution is skewed and can never be negative
- Two measures of degrees of freedom
  - df_1 = degrees of freedom for numerator
  - df_2 = degrees of freedom for denominator

Speaker notes

The F distribution appears many times in Statistics when you are comparing two different measures of variation. If it is close to one, that indicates that the two measures of variation are comparable. If it is a lot larger than one, that indicates that one source of variation (typically explained variation) is much larger.

Graph of the F distribution



Simon, 2024-10-22

Speaker notes

Here is a picture of the F distribution.

The F-test

- $H_0: \ \mu_1 = \mu_2 = \ldots = \mu_k$
- $H_1: \ \mu_i \neq \mu_j$ for some i, j
  - Compute F = MS(Between) / MS(Within)
  - Accept H0 if the F < $F(1-\alpha, k-1, N-k)$
- Only reject H_0 for large values
  - Note: the f-ratio can never be negative

Speaker notes

You only reject the null hypothesis for large values of F. If the variation between groups is about equal to the variation within groups or if it is much smaller than the variation within groups, you should accept the null hypothesis.

The p-value for the F test

- p-value = P[F(k-1, N-k) > F]
- Accept H_0 if p-value > α.

Speaker notes

R will compute a p-value for you and you compare the p-value to alpha. With a large p-value (greater than alpha), you should accept the null hypothesis. With a small p-value (less than or equal to alpha), you should reject the null hypothesis.

Can you use analysis of variance when you have two groups?

- Yes
  - $F = T^2$
- Some prefer the t-test with two groups
  - Simpler
- Some prefer analysis of variance
  - Generalizes to more complex settings (risk adjustment, interactions)

Speaker notes

While I specified analysis of variance as an approach that you use for three or more groups, it can be used when you only have two groups. The F statistic in analysis of variance is equal to the square of the T statistic in the two-sample t-test.

You will find that some researchers will only use the two sample t-test when there are two groups. Their rationale, in most cases, is that the two-sample t-test is the simpler approach and simpler is always better.

Other researchers, however, will adopt the analysis of variance framework at times. The analysis of variance framework allows you to consider more complex settings such as risk adjustment, and examination of interactions.

For most researchers, it is six of one and a half dozen of the other. Just be careful to never claim that analysis of variance works ONLY with three or more groups.

How does analysis of variance compare to the analysis of variance table in linear regression

- Linear regression SSR is comparable to analysis of variance SSB
  - Both represent explained variation
- Linear regression SSE is comparable to analysis of variance SSE
  - Both represent unexplained variation

Speaker notes

You’ve probably been wondering about the similarity of wording between the analysis of variance model and the analysis of variance table in linear regression. Analysis of variance is used when your independent variable is categorical and linear regression is used when your independent variable is continuous. Otherwise, they are pretty much the same thing. The regression sum of squares in linear regression and the between sum of squares in analysis of variance both represent explained variation. The error sum of squares in linear regression and the error sum of squares in analysis of variance both represent unexplained variation.

R code for the F test

```
       term df sumsq meansq statistic     p.value
1 factor(g)  2   336  168.0  11.58621 0.004337887
2 Residuals  8   116   14.5        NA          NA
```

Speaker notes

Because the F-ratio is large and the p-value is less than 0.05, you should reject H_0 and conclude that there is a difference among the means.

R-squared

- $R^2 = \frac{SS(Between)}{SS(Total)}$
  - Proportion of variation explained by groups

```
     logLik      AIC      BIC deviance nobs r.squared
1 -28.56465 65.12929 66.72087      116   11 0.7433628
```

Speaker notes

Approximately 74% of the variation in measurements can be accounted for by the grouping.

Break #2

- What you have learned
  - The F-test
- What's coming next
  - Assumptions

Population model

- There are k populations
  - $Y_{11}, Y_{12}, . \quad . \quad . , Y_{1N_1}$
  - $Y_{21}, Y_{22}, \ldots, Y_{2N_2}$
  - $\ldots$
  - $Y_{k1}, Y_{k2}, . \ . \ . , Y_{kN_k}$

Speaker notes

The population model in analysis of variance is effectively a description of k populations. The are N1 observations in the first population, N2 in the second population and Nk in the kth population. Notice the change in the spacing of the dot-dot-dot. This emphasizes that the k populations may not be all the same size.

Sampling from the population

- Select k samples
  - $Y_{11}, Y_{12}, \ldots, Y_{1n_1}$
  - $Y_{21}, Y_{22}, \ldots, Y_{2n_2}$
  - $\ldots$
  - $Y_{k1}, Y_{k2}, \ldots, Y_{kn_k}$

Speaker notes

Since N1, N2, etc. are denoted by capital letters, that reflects the fact that the size of the populations are often quite large–too large to measure every patient in each population. Instead you choose a smaller number of observations. Denote n1, n2, etc. as the size of the samples. Again, the spacing is slightly different because the size of the k samples do not have to be exactly equal.

Assumptions about the populations

- $Y_{ij} = \mu_i + \epsilon_{ij}$
  - ϵ_{ij} should be
    - Normal
    - Homogenous (equal variance)
    - Independent

Speaker notes

The three assumptions in analysis of variance are normality, homogeneity, and independence. Violations of the assumptions would be characterized as non-normality, heterogeneity, and lack of independence.

Residuals in analysis of variance

- In linear regression
  - $e_i = Y_i - \hat{Y}_i$
  - Residual = Observed - Predicted
- In analysis of variance
  - $e_{ij} = Y_{ij} - \bar{Y}_i$
  - Residual = Observed - Predicted
- Behavior of e_{ij} hints at assumptions on ϵ_{ij}

Speaker notes

You can compute residuals in analysis of variance much in the same way as you computed residuals in linear regression. In analysis of variance, the predicted values are the means in each group. So, in analysis of variance, you look at the deviations of the observed values from the group means, rather than deviations of the observed values from the trend line. But this is a fairly minor distinction. Always think of residuals as the difference between what you observed and what you predicted.

Important caveats

- Important notes
  - Check normality only in the residuals
  - Non-normality is less of a concern for large sample sizes
  - Unequal sample sizes is not a violation of assumptions

Speaker notes

The assumptions for analysis of variance are pretty much identical to the assumptions of the independent samples t-test and not all that different from the assumptions of linear regression. You need to assume normality, though it is normality of the residuals that you care about. The original outcome variable could be bimodal or trimodal or worse, just because the means are unequal. You are assuming that each group has a normal distribution, but the means of each normal distributions can be (and often are) quite different.

One more important note. You do not have to have exactly the same number of observations in each group. Unequal sample sizes is not a violation of assumptions.

Influential data points

- Less emphasis in analysis of variance
  - Impossible to have an outlier in the independent variable when it is categorical
- Studentized residuals may still be worth examining.

Speaker notes

In the modules on linear regression, you saw several measures of influence, leverage, studentized residuals, and Cook's distance. Leverage is a measure of how extreme an outlier is in the independent variable. When the independent variable is categorical, you almost never see an outlier in that dimension. It can happen when the first and second groups have several hundred patients and the third group only has five or ten patients. Such extreme imbalances are uncommon. Because of this, you really don't need to examine leverage. Cook's distance is not too critical either. There might be some value in looking at the studentized residual.

Actually, a simple set of boxplots works just as well as the studentized residual in identifying extreme values in the outcome measure.

How to check for non-normality

- Boxplots
  - Look for evidence of skewness or outliers

Speaker notes

Boxplots are very helpful for identifying skewness or outliers. Look for asymmetry. If the median is closer to one quartile than the other, that is an indication of skewness. If one whisker in the boxplot is a lot longer on one side than the other, that is also an indication of skewness.

Boxplots will highlight individual values that might be considered outliers.

How to check for heterogeneity

- Boxplots
  - Look for one box that is much wider than another
- Descriptive statistics
  - Look for one standard deviation much larger than another

Speaker notes

Boxplots are also useful for identifying heterogeneity (unequal variances). The relative size of the boxes in a boxplot help here. Don’t overinterpret small discrepancies. You shouldn’t really worry much about unequal variances unless the box in one boxplot two or three times wider that the box in another boxplot.

Likewise, look at the standard deviations in each group. Is one standard deviation a lot larger than another? Again, only worry if the disparity is large (two or three fold differences).

How to check for independence

- Qualitative assessment of how data was collected

Speaker notes

Although there are some formal tests for independence, these almost never apply. Do not use the runs test or the Durbin-Watson test unless there is a rigid temporal sequence in how the data was collected.

Instead, examine the assumption of independence by looking at details on how the data was collected.

Optional: analysis of residuals

- Normal probability plot (QQ plot) of residuals
- Histogram of residuals
- Plot predicted values versus residuals.

Speaker notes

While the boxplots are often quite informative, you might also want to look at various plots of the residuals.

The normal probability plot and the histogram help identify non-normality.

A plot of predicted values versus residuals helps identify heterogeneity. Recall that the predicted value in analysis of variance is just the group means. Be on the lookout for a fairly common pattern where groups with larger sample means also exhibit greater variation.

Break #3

- What you have learned
  - Assumptions
- What’s coming next
  - Confidence intervals

Multiple comparisons issue

- Type I error: rejecting the null hypothesis when the null hypothesis is true.
  - Multiple simultaneous hypotheses increase the Type I error rate.
- E_1 = Type I error for Hypothesis 1
- E_2 = Type I error for Hypothesis 2
  - $P[E_1 \cup E_2] = P[E_1] + P[E_2] - P[E_1 \cap E_2]$
  - $P[E_1 \cup E_2] \leq P[E_1] + P[E_2]$
  - $P[E_1 \cup E_2] \leq 2\alpha$

Speaker notes

Recall that a Type I error is rejecting the null hypothesis when the null hypothesis is true. In analysis of variance, there are several types of deviations that you might want to investigate. Let’s look at this from a theoretical perspective. Let’s suppose there are two alternative hypotheses H1 and H2 and you will reject the null hypothesis if either one of the alterantives is true. A bit of math will show that if you have an alpha level for each of the two hypotheses, then your chance of making a Type II error almost doubles. You thought you were controlling the Type I error rate at 5% but you let the Type I error rate creep up to almost 10%.

Bonferroni adjustment

- For m hypotheses
  - $P[E_1 \cup \ldots \cup E_m] \leq m\alpha$
- Test each hypothesis at α/m
  - Preserves overall Type I error rate
- Example, 3 simultaneous hypotheses
  - Reject H0 if p-value < 0.05/3 or 0.0167

Speaker notes

For multiple hypotheses, it gets worse. With m hypotheses, your Type I error rate gets multiplied by m. Now it isn't that bad in reality, but it is close. With 3 hypotheses, a Type I error rate that holds individually at 5% can be almost as bad as 15%. With 10 hypotheses, it can be almost 50%.

The solution is to test each individual hypothesis at a smaller level. If you have three ways that the null hypothesis might be rejected. Test each at a 0.05/3 or 0.0167 Type I error rate.

Tukey post hoc tests

Speaker notes

A similar approach is known as the Tukey post hoc test. This is based on a statistic known as the studentized range. The details are not important, but it helps you identify for which values of i and j, the groups differ. And even though there are k (k-1)/2 comparisons, the Tukey post hoc test adjusts the individual Type I error rates so that the overall Type I error rate is still less than alpha.

Tukey post hoc test for k=3

- $H_1: \ \mu_i \neq \mu_j$ for some i, j
- There are 3 sub-hypotheses
  - $H_{1a}: \ \mu_1 \neq \mu_2$ and/or
  - $H_{1b}: \ \mu_1 \neq \mu_3$ and/or
  - $H_{1c}: \ \mu_2 \neq \mu_3$
    - Does the confidence interval exclude zero
    - Is the adjusted p-value less than alpha?

Speaker notes

If there are three groups, the Tukey post hoc test examines three sub-hypotheses: mu1 not equal to mu2, mu1 not equal to mu3, and/or mu2 not equal to mu3.

If the confidence interval for mu1-mu2 excludes zero, then you would conclude that mu1 is not equal to mu2. Likewise if the confidence interval for mu1-mu3 excludes zero, then you would conclude that mu1 is not equal to mu3. If the confidence interval for mu2-mu3 excludes zero, then you would claim that mu2 is not equal to mu3.

Equivalently, you could look at the adjusted p-value. If the adjusted p-value is less than alpha for a particular pair, conclude that the population means for that pair are different.

Tukey post hoc test for k=4

- $H_1: \ \mu_i \neq \mu_j$ for some i, j
- There are 6 sub-hypotheses
  - $H_{1a}: \ \mu_1 \neq \mu_2$
  - $H_{1b}: \ \mu_1 \neq \mu_3$
  - $H_{1c}: \ \mu_1 \neq \mu_4$
  - $H_{1d}: \ \mu_2 \neq \mu_3$
  - $H_{1e}: \ \mu_2 \neq \mu_4$
  - $H_{1f}: \ \mu_3 \neq \mu_4$

Speaker notes

If there are four groups, then there are six possible pairwise comparisons. Compute six different confidence intervals and then see which ones include or exclude the value of zero.

Alternatives to Tukey post hoc tests, 1

- Bonferroni adjustment
  - Works for unequal sample sizes per group
  - Works for unequal variances
- Dunnett's test
  - Multiple treatments versus a single control
- Scheffe's test
  - Works for complex comparison
    - Example $\mu_1 \ vs. \ \frac{\mu_2+\mu_3+\mu_4}{3}$

Speaker notes

There are several good alternatives to the Tukey post hoc test. You won’t have to run any of these in your programming assignment, but it helps to mention their names in case you see reference to them in a published research paper.

The Bonferroni adjustment, mentioned earlier is not quite as good as the Tukey post hoc test, but it still works even with unequal sample sizes and even with unequal variances.

The Dunnett’s test is helpful when you have multiple treatment and you want to compare each treatment to a control group, but you are not interested in comparisons within each treatment. In this setting, you get a bit more precision and power with Dunnett’s test compared to the Tukey post hoc test.

Finally there is Scheffe’s test which works for more complex comparisons such as whether the first group mean differs from an average of the second, third, and fourth group means.

Controversies over Tukey/Bonferroni adjustment

- Criticisms that extend beyond one factor analysis of variance.
- Increases Type II errors
- Impractical for large values of m

Speaker notes

Although most researchers like using a Tukey or Bonferroni adjustment in analysis of variance, there is a vocal minority who argue that you should not use them, either in one factor analysis of variance or in more complex settings. These researchers suggest that no adjustment is better than Tukey or Bonferroni adjustments because while Tukey and Bonferroni do succeed in controlling the Type I error rate, they pay a high price (too high a price in some people's minds) in an increase in the Type II error rates.

The other problem is that these adjustments are impractical for very large numbers of comparisons, such as you might see in many genomics studies.

There is something to be said for these critiques, but you will probably have less resistance in any peer review of your work if you do try to control the Type I error rate with Tukey or Bonferroni.

Ambiguities in Tukey post hoc test results

- One possible outcome
  - Conclude that $\mu_1 = \mu_2$
  - Conclude that $\mu_2 = \mu_3$
  - Conclude that $\mu_1 \neq \mu_3$

Speaker notes

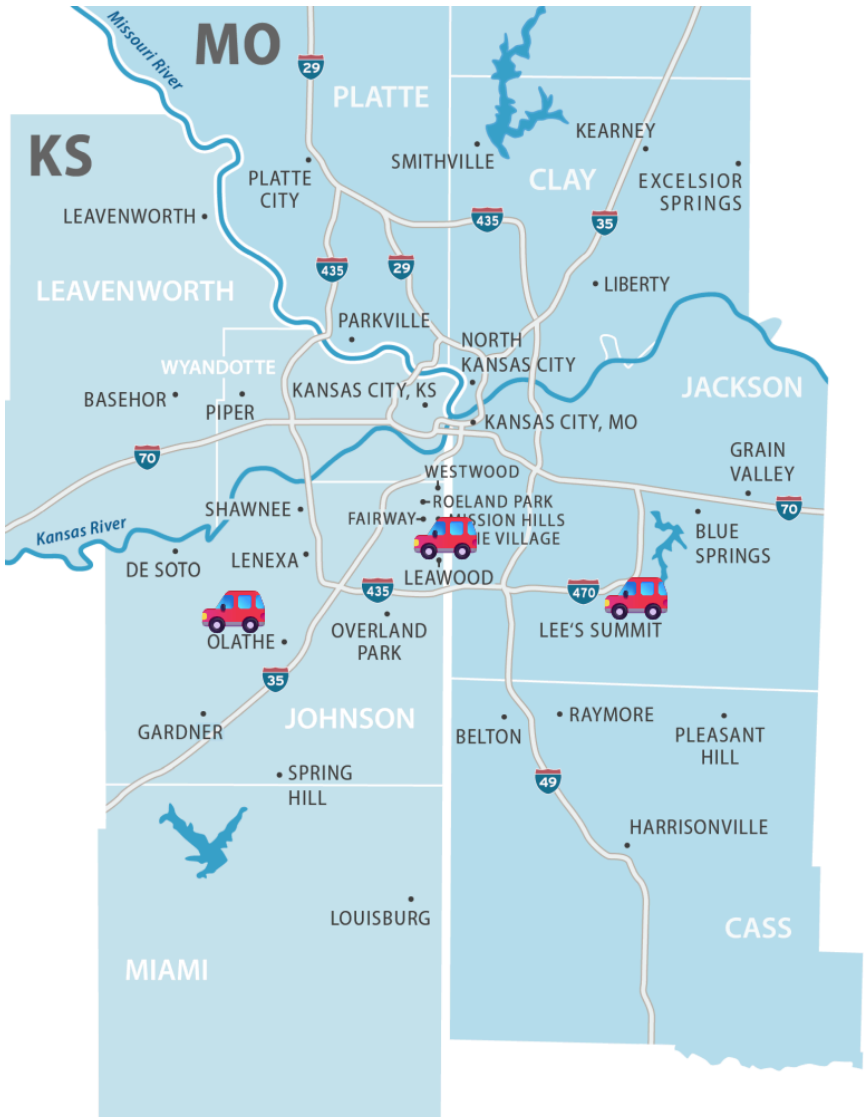
When you have three or more groups, the results of your Tukey post hoc tests can sometimes be ambiguous. You might, for example, conclude that mu1 equals mu2 becaose the confidence interval for mu1-mu2 includes the value of zero. And you might also conclude that mu2 dquals mu3 because the confidence interval for mu2 minus mu3 includes the value of zero.

But, it could turn out that you would have the confidence interval for mu1-mu3 excluding the value of zero which would lead you to conclude that mu1 is not equal to mu3.

How could you have mu1 equal to mu2 and mu2 equal to mu3 but not have mu1 equal to mu3?

It turns out thatuse of a sample means that you make statements about whether xbar1-xbar2, xbar2-xbar3, and xbar1-xbar3 are close to zero or not. And depending on where the means lie, it is possible for xbar1 to be close to xbar2, xbar2 be close to xbar3 but xbar1 NOT be close to xbar3.

Geography analogy



Speaker notes

Here’s a map that illsutrates this. Steve Simon lives in Leawood. That’s the middle car on this map. His son, Nick, used to live with him, but has moved to his own place in Lee’s Summit. That’s the easternmost car on this map. Nick gets his car serviced in Olathe. That’s the westernmost car on the map.

Now Leawood and Olathe are close to each other. When Nick lived in Leawood, he was close to Olathe. When he moved to Lee’s summit, he was still close to Steve’s home in Leawood, but now he was far away from his car repair shop in Olathe.

So Olathe and Leawood are close. Leawood and Lee’s Summit are close. That does not mean that Olathe and Lee’s Summit are close.

That’s an analogy to the unexpected findings that you will see at times with the Tukey post hoc test.

Break #4

- What you have learned
  - Confidence intervals
- What’s coming next
  - R code for analysis of variance

Live demonstration, part 1

The following demonstration uses the wolf-river-pollution dataset. Refer to the data dictionary found on my github site.

You can also find the R code on my github site.

Break #5

- What you have learned
  - R code for analysis of variance
- What’s coming next
  - Sample size justification

Using the power.anova.test function to estimate sample size

Speaker notes

Estimating sample size is a bit trickier for analysis of variance. The power.anova.test does a pretty good job of handling this case.

What if n is outside your budget?

- Increase between.var
- Increase Type I error rate (sig.level)
- Increase Type II error rate (decrease power)
- Decrease number of groups

Speaker notes

You may find that your first shot at estimating the sample size causes a panic in your reserach team. There are some changes that you might consider.

First, look at the various group means that are used to calculate the between variance. Maybe you are hoping to detect a really small difference among these means. Can you live with a research design that only has good power for a slightly wide range of group means?

You might consider increasing the Type I error rate. There is no good reason other than tradition to hold the Type I error rate at 0.05 and you will find that a Type I error rate of 0.10 will require a slightly smaller sample size. Now, realistically, it is hard to publish research that deviates from the widely accepted practice of keep the alpha level at 0.05.

You could increase the Type II error rate (or equivalently decrease the power). Using a Type II error rate of 0.10 *(or power equals 0.90) is a good starting point. Perhaps a Type II error rate of 0.20 (power equals 0.80) might be tolerable. Don’t try to go more extreme than this. I might not mind, but the are others who do kick up a fuss for any power less than 0.80.

You can also decrease the sample size markedly by decreasing the number of groups. Can you live with three groups instead of six? You’ll cut your sample size by more than half, actually, because the F distribution behaves a lot nicer with fewer groups.

Break #6

- What you have learned
  - Sample size justification
- What’s coming next
  - R code for sample size justification

Live demonstration, part 2

This is a continuation of simon-5501-10-demo.

Break #7

- What you have learned
  - R code for sample size justification
- What's coming next
  - Your homework

Your homework

The programming assignment is available through Canvas, but it actually links to a page on my github site.

Summary

- What you have learned
  - The analysis of variance model
  - The F-test
  - Assumptions
  - Confidence intervals
  - R code for analysis of variance
  - Sample size justification
  - R code for sample size justification
  - Your homework